higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 4

(6021034)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Draw with a pencil and ruler.
5. Do NOT colour drawings with liquid ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Method of increasing the engine compression ratio:

A Filling the engine with more oil
B Installing a thinner cylinder head gasket
C Fitting a hollow-head piston
D Installing double cylinder head gaskets

1.1.2 Which ONE of the following components is NOT found in an automatic gearbox:

A Cotter
B Brake belts
C Gears
D Governors

1.1.3 During the building up of a magnetic field in an ignition coil the contact points are …

A completely open.
B completely closed.
C partially open.
D partially closed.

1.1.4 Which ONE of the following questions must be asked when beginning to diagnose a conventional final drive:

A Is it in first, second gear or overdrive?
B When does the noise occur?
C Does it only occur when a friend drives?
D Does it happen in reverse gear?

1.1.5 Which ONE of the following is NOT a method of fuel injection:

A Throttle-body injection
B Carburettor injection
C Port injection
D Indirect injection

(5 × 1) (5)

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1.2 Figure 1 below, is a sketch of a sectioned view of the gears inside a differential housing.

Label the parts marked (A to H) by writing only the name of each part next to the letter (A–H) in the ANSWER BOOK.

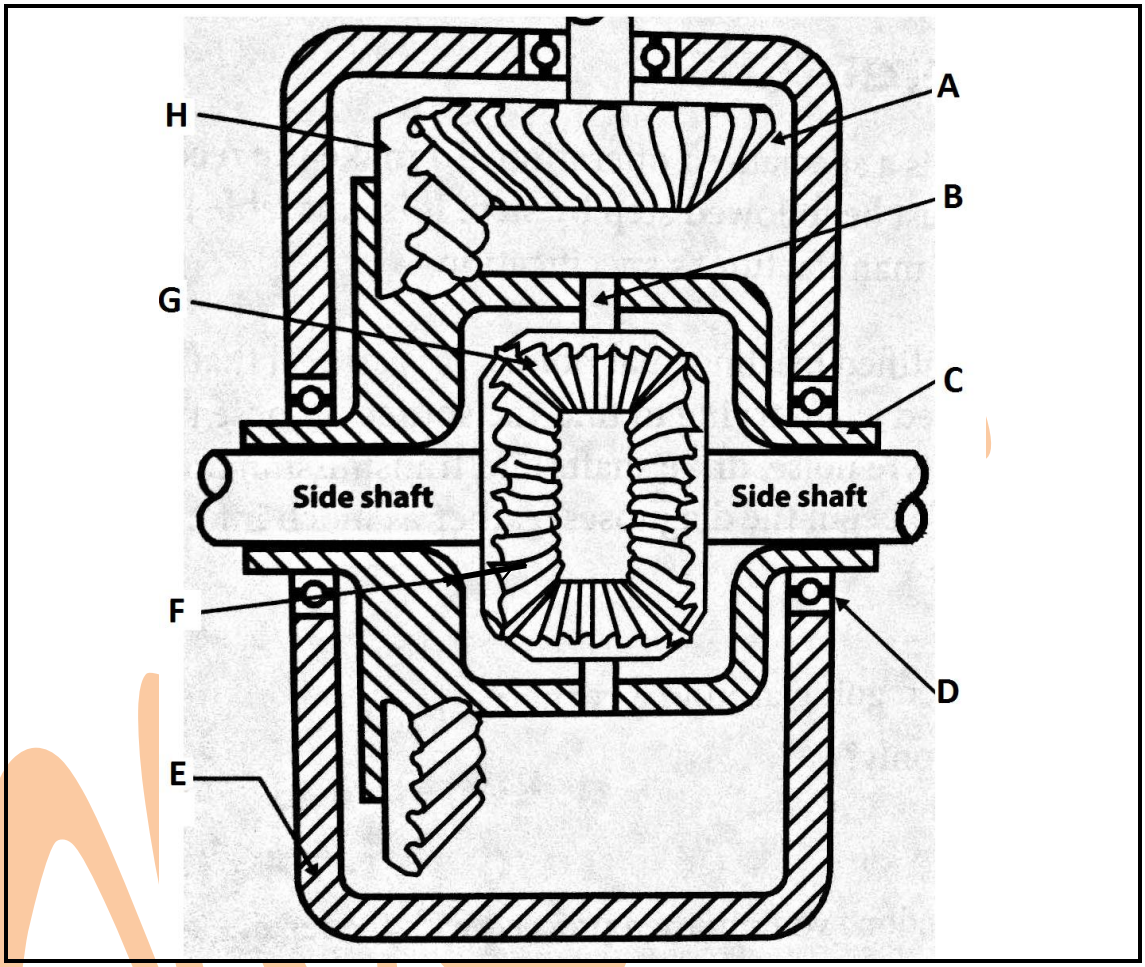


FIGURE 1 (8 × 1) (8)

1.3 Compare the disadvantages of a power-steering system to its advantages. (7)
[20]

QUESTION 2

2.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–M) next to the question number (2.1.1–2.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1.1	Backlash	A	kinetic friction and static friction are exposed
2.1.2	Passive sensor	B	too much torque will overcome the friction
2.1.3	Active sensor	C	assists with a movement
2.1.4	Vehicle-speed sensor	D	do not produce their own voltage
2.1.5	Solenoid valve	E	is placed on the tyre and road surface due to the sideways movement
2.1.6	Airflow sensor	F	tested very much the same way as switches
2.1.7	Augmenting	G	device that monitors the flow of air into the engine
2.1.8	Braking forces	H	sends messages to the computer which tells the transmission how to perform
2.1.9	Acceleration forces	I	voltage produced is very low – under 1 volt
2.1.10	Cornering forces	J	main function is to indicate that the transmission is in neutral
		K	spark plugs are shorted to the ground
		L	clearance between meshed gears
		M	converts electrical energy into mechanical energy

(10 × 1) (10)

2.2 Discuss the advantages and disadvantages of *epicycle gearing* over *conventional gearing*. (3 + 3) (6)

2.3 Name FOUR elements a single epicycle gear train comprises. (4)

[20]

QUESTION 3

3.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.1.1–3.1.5) in the ANSWER BOOK.

3.1.1 The main journal turns twice the speed of the big-end journal on the crank shaft.

3.1.2 Too much torque will overcome friction and the vehicle wheels will spin.

3.1.3 The coefficient of friction is determined by the 'stickiness' between the two surfaces.

3.1.4 The hybrid electrical car used a 6 V and 12 V conventional batteries.

3.1.5 Auxiliary equipment is powered by a HV battery.

(5 × 1) (5)

3.2 Explain the difference between EACH of the following:

3.2.1 Swept volume and volumetric efficiency

3.2.2 Solenoid and sensor

3.2.3 Mechanical efficiency and thermal efficiency

3.2.4 Static crankshaft balancing and dynamic crankshaft balancing

3.2.5 Compression test and cylinder-balance test

(5 × 2) (10)

3.3 Explain the operation of a single-speed double-reduction differential. (5)

[20]

QUESTION 4

4.1 Complete the following paragraph by using the words in the list below. Write only the missing word next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

power; higher; less; increases; decreases; volumetric; pressure

As the altitude increases, the atmospheric pressure (4.1.1) … Less air is being admitted to the cylinder, resulting in lower (4.1.2) … efficiency and therefore the mean effective (4.1.3) … acting on the piston during power stroke is (4.1.4) … As the engine operates at a (4.1.5) … altitude the loss of power is inevitable. (5 × 1) (5)

4.2 Compare the advantages of a compression ignition (CI) engine to its' disadvantages. (5 + 5) (10)

4.3 Explain the term *thermal efficiency*. (5)

[20]

QUESTION 5

5.1 Explain the concept of *air standard efficiency.* (5)

5.2 State FIVE methods to increase the compression ratio of an engine. (5)

5.3 Explain how to perform a cylinder-leakage test on a petrol engine. (5)

5.4 Rearrange the following steps to replace faulty parts on an electronic ignition system in the correct sequence. Write the correct step next to the question number (5.4.1–5.4.5) in the ANSWER BOOK.

5.4.1 Test the system to ensure that the integrity of the wiring is in a serviceable condition.

5.4.2 Remove and replace the faulty electronic ignition system component with parts that meet the manufacturer's specifications.

5.4.3 Investigate and apply safe working practices throughout the task.

5.4.4 Confirm if the electronic ignition system operates according to the manufacturer's specifications.

5.4.5 Check all adjustments and, where necessary, correct to comply with the manufacturer's specifications.

(5 × 1) (5)

[20]

TOTAL: 100

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